Experiment No: 04

Name of The Experiment: Study of Series and parallel circuit.

Objectives:

a) To connects two types of connection circuits (parallel & series).
b) To measure the Voltages and current across the resistors using the Multi-meter device.
c) To analysis the result and performance of the circuits.

Theory:

THE SERIES CIRCUIT

A Series Circuit or ''series-connected circuit'' is a circuit having just one current path. Thus, Fig 4(a) is an example of a ''series circuit'' in which a battery of constant potential difference V volts, and three resistances, are all connected ''in series.''

Since a series circuit has just one current path, it follows that all the components in a series circuit carry the same current I, a fact evident from inspection of Fig 4(a). The current, I is assumed to be a flow of positive charge, and thus flows out of the positive terminal of the battery and around through the external circuit, reentering the battery at the negative terminal. This is indicated by the arrows in Fig 4(a).

In a series circuit, the TOTAL resistance, RT, that the battery sees is equal to the SUM of the individual resistances. Thus, in the particular case of Fig 4(a) the battery sees a total resistance, $R_T = R_1 + R_2 + R_3$, while in the general case of ''n'' resistances connected in series the battery sees a total resistance of....

$R_T = R_1 + R_2 + R_3 + \ldots . R_n$

By Ohm's law, it follows that the current I in a series circuit is equal to

$$I = \frac{V}{R_T} = \frac{V}{R_1 + R_2 + \cdots + R_n}$$

In a series circuit, the applied voltage is equal to the sum of the voltage drops.

$V_T = V_1 + V_2 + V_3$

Scanned with CamScanner

THE PARALLEL CIRCUIT

A PARALLEL circuit is one in which the battery current divides into a number of "parallel paths." This is shown in Fig 4(b), in which a battery, of constant V volts, delivers a current of I amperes to a load consisting of any number of n resistances connected "in parallel."

The currents in the individual resistances are called the "branch currents," and the battery current I is often called the "line current." From inspection of Fig 4(b) we see that, in a parallel circuit, the battery current I is equal to the sum of the branch currents.

$$I = I_1 + I_2 + I_3 + \cdots + I_n$$

If the battery voltage V is applied equally to all n resistances; that is, the same voltage V is applied to all the parallel branches. Hence, by Ohm's law, the individual branch currents in Fig 4(a) have the values:

$$I_1 = V/R_1, \qquad I_2 = V/R_2, \ldots, I_n = V/R_n$$

Then, we have:

$$I = V\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}\right)$$

Now let RT be the total resistance as seen by the battery in Fig 4(b). Then, by Ohm's law, it has to be true that:

$$I = \frac{V}{R_T}$$

Since the left-hand sides of the last two equations are equal, the two right hand sides are also equal. Setting the two right-hand sides equal, then canceling the Vs, gives

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}$$

Scanned with CamScanner

Theoretical Diagram:

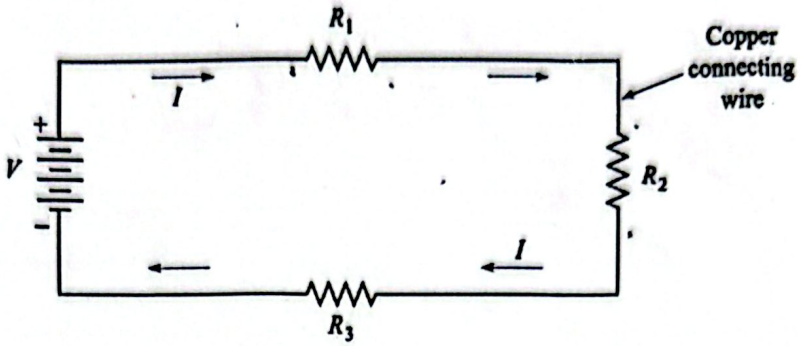


Fig 4(a): A Series Circuit

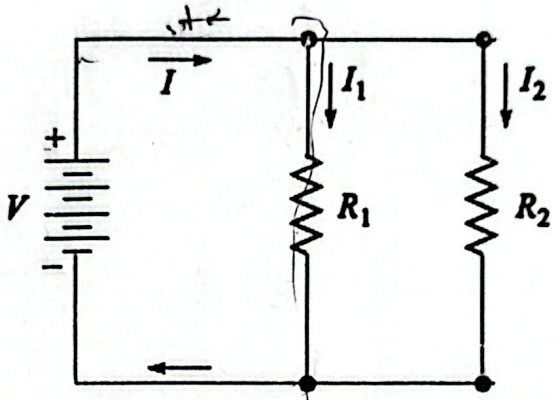


Fig 4(b): A Parallel Circuit

Apparatus:

a) DC power supply
b) Breadboard
c) Resistors of various values
d) Voltmeter
e) Ammeter
f) Connecting wires

Procedure:

a) Connect all components on the bread board according to the circuit of Fig 4(a) on the bread board.
b) Turn on the dc power supply.
c) Measure the voltage and current using voltmeter and ammeter respectively.
d) Repeat the above step for some times.

Scanned with CamScanner

Experimental Table:

Data Table 01: A Series Circuit

OBS	R_1 (Ω)	R_2 (Ω)	R_3 (Ω)	V_1(V)		V_2(V)		V_3(V)		R_S(Ω)	
				Cal	Mea	Cal	Mea	Cal	Mea	Cal	Mea
1.	470	2200	1000								
2.											
3.											

Data Table 02: A Parallel Circuit

OBS	R_1(Ω)	R_2(Ω)	I_1(A)		I_2(A)		R_p(Ω)	
			Cal	Mea	Cal	Mea	Cal	Mea
1.								
2.								
3.								

Calculations:

A Series Circuit:

OBS 1:

$R_s = R_1 + R_2 + R_3$ .....[calculated value]

$V1 = \frac{V \times R1}{R1+R2+R3}$ (Volt) .....[calculated value]

$V2 = \frac{V \times R2}{R1+R2+R3}$ (Volt) .....[calculated value]

$V3 = \frac{V \times R3}{R1+R2+R3}$ (Volt) .....[calculated value]

$Rs = \frac{V}{I} = \frac{V1+V2+V3}{I}$ .....[measured value]

Scanned with CamScanner

OBS 2:

$R_S = R_1 + R_2 + R_3$ .....[calculated value]

$V1 = \frac{V \times R1}{R1+R2+R3}$ (Volt) .....[calculated value]

$V2 = \frac{V \times R2}{R1+R2+R3}$ (Volt) .....[calculated value]

$V3 = \frac{V \times R3}{R1+R2+R3}$ (Volt) .....[calculated value]

$R_S = \frac{V}{I} = \frac{V\,1+V2+V3}{I}$.....[measured value]

OBS 3:

$R_S = R_1 + R_2 + R_3$ .....[calculated value]

$V1 = \frac{V \times R1}{R1+R2+R3}$ (Volt) .....[calculated value]

$V2 = \frac{V \times R2}{R1+R2+R3}$ (Volt) .....[calculated value]

$V3 = \frac{V \times R3}{R1+R2+R3}$ (Volt) .....[calculated value]

$R_S = \frac{V}{I} = \frac{V\,1+V2+V3}{I}$.....[measured value]

A Parallel Circuit:

OBS 1:

$R_P = = \frac{R1 \times R2}{R1+R2}$(ohm) .....[calculated value]

$I_1 = \frac{I \times R2}{R1+R2}$ (A) .....[calculated value]

$_2 = \frac{I \times R1}{R1+R2}$ (A) .....[calculated value]

$R_P = \frac{V}{I} = \frac{V}{I1+I}$(ohm).....[measured value]

Scanned with CamScanner

OBS 2:

$R_P = = \frac{R1 \times R2}{R1+R2}$ (ohm) .....[calculated value]

$I_1 = \frac{I \times R2}{R1+R2}$ (A) .....[calculated value]

$I_2 = \frac{I \times R1}{R1+R2}$ (A) .....[calculated value]

$R_P = \frac{V}{I} = \frac{V}{I1+I2}$(ohm).....[measured value]

OBS 3:

$R_P = = \frac{R1 \times R2}{R1+R2}$ (ohm) .....[calculated value]

$I_1 = \frac{I \times R2}{R1+R2}$ (A) .....[calculated value]

$I_2 = \frac{I \times R1}{R1+R2}$ (A) .....[calculated value]

$R_P = \frac{V}{I} = \frac{V}{I1+I}$(ohm).....[measured value]

Results:

A Series Circuit:

a) OBS 1: Measured Value of resistor……-- Calculated Value of resistor …..

b) OBS 2: Measured Value of resistor……-- Calculated Value of resistor …..

c) OBS 3: Measured Value of resistor……-- Calculated Value of resistor …..

A Parallel Circuit:

a) OBS 1: Measured Value of resistor……-- Calculated Value of resistor …..

b) OBS 2: Measured Value of resistor……-- Calculated Value of resistor …..

c) OBS 3: Measured Value of resistor……-- Calculated Value of resistor …..

Scanned with CamScanner

Discussion:

The observed values and calculated values are nearly same. The difference between the is for instrumental and observational error. Neglecting this error Series and parallel circuits are verified successfully.

N.B: Fluctuations in temperature, humidity, and other environmental conditions can affect the performance of components and measurement instruments.

Conclusion

References

Scanned with CamScanner